

Yichao Jin

PhD in Economics, UT Dallas

Email: yichao.jin@utdallas.edu | Phone: (682) 266-4185

Dallas, TX

July 8, 2026

Dr. Zachary Butzin-Dozier

Assistant Professor of Pediatrics (Clinical Informatics)

and of Medicine (Computational Medicine)

Stanford University School of Medicine

Re: Postdoctoral Position in Causal Inference and EHR Research

Dear Dr. Butzin-Dozier,

I am writing to apply for the postdoctoral position in your lab advertised through the Stanford Division of Clinical Informatics. I recently completed my PhD in Economics at UT Dallas with a focus on causal inference and machine learning in health decision-making, and I believe my methodological background aligns closely with the research program you are building.

My recent research developed an Empirical-Frontier Regularization (EFR) framework that integrates machine learning with behavioral consistency constraints -- essentially a method to audit and calibrate LLM-generated health preference predictions against empirical benchmarks. This work sits at the intersection of causal inference, machine learning, and observational health data, which is also the core of your research on targeted machine learning for EHR-based causal effect estimation in Long COVID.

Your work applying targeted machine learning to the N3C -- evaluating protective interventions like SSRIs and metformin against Long COVID -- is a natural ground to extend my methods. The EFR framework's logic of imposing behavioral and statistical constraints on data-driven predictions translates directly to the problem of drawing reliable causal inferences from complex, high-dimensional EHR data. I am particularly interested in exploring how targeted learning and super learner ensembles (which I have used in my own work) can be combined with regularization approaches to improve the robustness of treatment effect estimates in heterogeneous patient populations.

Beyond my methodological work, I have extensive experience with discrete choice experiments and health preference modeling, which I see as complementary to EHR-based research -- DCE data captures patient preferences that are invisible in clinical records, and linking these two data types could deepen your program's understanding of patient-centered outcomes. I also hold a Master's in Statistics and Data Analytics, giving me strong computational foundations in R and Python.

I am available to start as early as August 2026 and have current work authorization through my F-1 visa. My CV is attached. I would welcome the opportunity to discuss how my background might contribute to the research directions you are developing in your new lab.

Best regards,

Yichao Jin